

Since $AB=I$, By the Invertible Matrix Theorem and the equivalence of the statements K and a , we know that A is invertible.

By the defn. of inverse, there exists a $n \times n$ matrix A^{-1} s.t. $AA^{-1}=A^{-1}A=I_n$.

$$AB=I \Rightarrow A^{-1}(AB)=A^{-1}I \Rightarrow (A^{-1}A) \cdot B = A^{-1}I \quad (\text{Theorem 2(a)})$$

$$\Rightarrow I \cdot B = A^{-1}I \quad (\text{By the defn. of inverse}) \Rightarrow \boxed{B=A^{-1}} \quad (\text{Theorem 2(e)}) \quad \square$$

26) Explain why the columns of A^2 span $\mathbb{R}^n$ whenever the columns of A are linearly independent.

Solution: Given, the columns of A are linearly independent.
By the Invertible Matrix Theorem and the equivalence of the statements (e) and (a), we know that A is invertible. $A^2=A \cdot A$

By Theorem 6(b), A^2 is invertible.
By the Invertible Matrix Theorem and the equivalence of the statements (a) and (h), The columns of A^2 span $\mathbb{R}^n$. $\square$

27) Show that if AB is invertible, so is A .

Solution: By defn. of inverse, there exists a $n \times n$ matrix X s.t. $(AB)X=I$ (Theorem 2(a)). BX is an $n \times n$ matrix.

By the Invertible Matrix Theorem and the equivalence of the statements K and a , we know that A is invertible. $\square$

28) Show that if AB is invertible, so is B .

Solution: By defn. of inverse, there exists a $n \times n$ matrix X s.t. $X(AB)=I$
 $\Rightarrow (XA)B=I$ (Theorem 2(a)). XA is an $n \times n$ matrix.

By the Invertible Matrix Theorem and the equivalence of the statements (b) and (a), B is an invertible matrix. $\square$

29) If A is an $n \times n$ matrix and the equation $Ax=b$ has more than one solution for some b , then the transformation $x \mapsto Ax$ is not one-to-one. What else can you say about this transformation? Justify your answer.

Solution: By the Invertible Matrix Theorem and the equivalence of the statements f , i and a , we know that the linear transformation $x \mapsto Ax$ does not map $\mathbb{R}^n$ onto $\mathbb{R}^n$, and A is